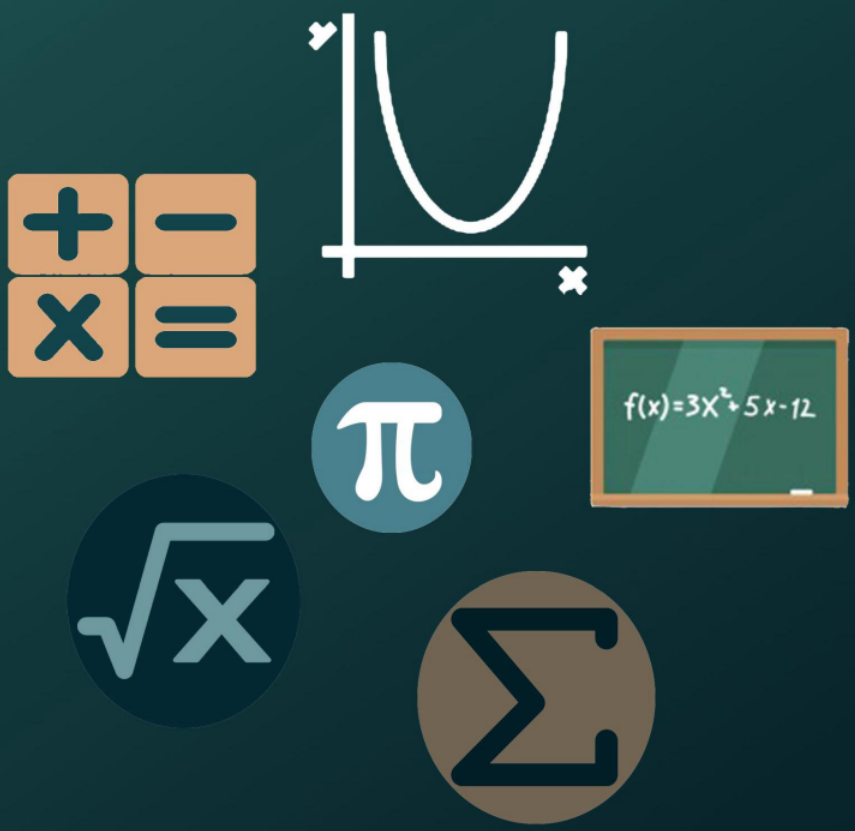
MATHEMATICS

Logarithm



SHEET

eSaral.com

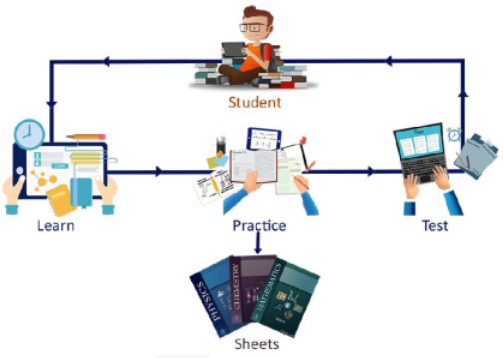
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 3
5.	Exercise 2	Page - 6
6.	Exercise 2A	Page - 8
7.	Answer Key	Page - 10

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Logarithm				
2	Basics of Logarithm				
3	Properties of Logarithm - Part 1	1 - 6	1, 2		1, 2
4	Properties of Logarithm - Part 2	7 - 17	3, 4, 5, 11, 14 - 22	1 - 6, 11 - 13	3, 4, 15, 16, 18 - 21
5	Logarithmic Equations	18 - 27	6, 12, 13, 23 - 27	7 - 10	5, 7 - 14, 17, 22 - 24
6	Logarithmic Graphs	28	8 - 10		
7	Antilogarithm	29			
8	Characteristic & Mantissa	30 - 34	7	14, 15	6

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Properties of Logarithm

Q1. The value of $7\log\frac{16}{15}+5\log\frac{25}{24}+3\log\frac{81}{80}$ simplifies to -

(A) log1 (B) log2
(C) log3 (D) log4

Q2. What is logarithm of $32\sqrt[5]{4}$ to the base $2\sqrt{2}$:

(A) $\left(\frac{5}{18}\right)$ (B) $\left(\frac{9}{5}\right)$
(C) $\left(\frac{18}{5}\right)$ (D) $\left(\frac{36}{5}\right)$

Q3. Find the value of

$2\log\frac{2}{5}+3\log\frac{25}{8}-\log\frac{625}{128}$

(A) 1 (B) 2
(C) 3 (D) 0

Q4. $\log_2\left(1+\frac{1}{2}\right)+\log_2\left(1+\frac{1}{3}\right)+.....+\log_2\left(1+\frac{1}{31}\right)$ simplifies to -

(A) 0 (B) 1
(C) 5 (D) 4

Q5. If $2\log(x+y)=3\log5+\log x+\log y$, then $\frac{(x+y)^2}{xy}$ is -

(A) 125 (B) 121
(C) 25 (D) 50

Q6. If $a^2+b^2=23ab$, then what is the value of $\log\frac{(a+b)}{5}$.

(A) $\frac{1}{2}(\log a+\log b)$ (B) $\frac{1}{2}(\log a-\log b)$
(C) $\frac{1}{2}\log a$ (D) $\frac{1}{2}\log b$

Q7. If $2^a=3$ and $9^b=4$ then value of (ab) is-

(A) 1 (B) 2
(C) 3 (D) 4

Q8. If $a=\log_{24}12$, $b=\log_{36}24$, $c=\log_{48}36$, then $\frac{1+abc}{bc}$ equals -

(A) 0 (B) 1
(C) 2 (D) 4

Q9. The value of the expression $\frac{2}{\log_4(2000)^6}+\frac{3}{\log_5(2000)^6}$ is -

(A) $\left(\frac{1}{3}\right)$ (B) $\left(\frac{1}{2}\right)$
(C) $\left(\frac{1}{6}\right)$ (D) $\left(\frac{2}{3}\right)$

Q10. What is the value of $\frac{\log_2 24}{\log_{96}2}-\frac{\log_2 192}{\log_{12}2}$.

(A) 1 (B) 2
(C) 3 (D) 0

Q11. If $\log_{10}2=0.3010$ and $\log_{10}3=0.4771$, then the value of $\log_5(512)$ is :

(A) $\frac{2709}{699}$ (B) $\frac{699}{2709}$
(C) $\frac{4771}{3010}$ (D) $\frac{3010}{699}$

Q12. Find value of

$$\frac{1}{1+\log_a(bc)}+\frac{1}{1+\log_b(ac)}+\frac{1}{1+\log_c(ab)}.$$

(A) 2 (B) 3
(C) 1 (D) 4

Q13. Find the value of
$(\log_3 4)(\log_4 5)(\log_5 6)(\log_6 7)(\log_7 8)(\log_8 9)$:
(A) 1 (B) 2
(C) 0 (D) 4

Q14. If $a^x = b$, $b^y = c$, $c^z = a$, find value of xyz :
(A) 3 (B) 1
(C) 5 (D) 4

Q15. Find the value of $49^{(1-\log_7 2)} + 5^{-\log_5 4}$.

(A) $\frac{24}{2}$ (B) $\frac{25}{3}$

(C) $\frac{21}{4}$ (D) $\frac{25}{2}$

Q16. The value of $3^{\log_4 5} + 4^{\log_5 3} - 5^{\log_4 3} - 3^{\log_5 4}$ is -
(A) 0 (B) 1
(C) 2 (D) none of these

Q17. Evaluate : $81^{1/\log_5 3} + 27^{\log_9 36} + 3^{4/\log_7 9}$.
(A) 890 (B) 980
(C) 800 (D) 811

Q18. Value of

$$\left(\frac{1}{\log_6(240)}+\frac{1}{\log_8(240)}+\frac{1}{\log_5(240)}\right) \text{ is :}$$

(A) 1 (B) 0
(C) 3 (D) 2

Logarithm Equation

Q19. If $\log_2\left(4+\log_3(x)\right)=3$, then sum of digits of x is -
(A) 3 (B) 6
(C) 9 (D) 18

Q20. Sum of all the solution(s) of the equation $\log_{10}(x) + \log_{10}(x+2) - \log_{10}(5x+4) = 0$ is-
(A) –1 (B) 3
(C) 4 (D) 5

Q21. The product of all the solutions of the equation $x^{1+\log_{10} x} = 100000x$ is-
(A) 10 (B) 10^5
(C) 10^{-5} (D) 1

Q22. If x_1 and x_2 are the roots of equation $e^{3/2}.x^{2\ln x} = x^4$, then the product of the roots of the equation is-
(A) e^2 (B) e
(C) $e^{3/2}$ (D) e^{-2}

Q23. If $\log_2(x^2+1) + \log_{13}(x^2+1) = \log_2(x^2+1)\log_{13}(x^2+1), (x \neq 0)$, then $\log_7(x^2+24)$ is equal to-
(A) 1 (B) 2
(C) 3 (D) 4

Q24. Find 'x' satisfying the equation
$4^{\log_{10} x+1} - 6^{\log_{10} x} - 2.3^{\log_{10} x^2+2} = 0$.

(A) $x = \frac{1}{10}$ (B) $x = \frac{1}{100}$

(C) $x = \frac{1}{1000}$ (D) 0

Q25. The value of the expression, $\log_4\left(\frac{x^2}{4}\right) - 2\log_4(4x^4)$ when $x = -2$ is -
(A) – 6 (B) – 5
(C) – 4 (D) meaningless

Q26. If $\log_a\left(1-\sqrt{1+x}\right) = \log_{a^2}\left(3-\sqrt{1+x}\right)$, then number of solutions of the equation is-
(A) 0 (B) 1
(C) 2 (D) infinitely many

Q27. The number of solution(s) of $\sqrt{\log_3(3x^2).\log_9(81x)} = \log_9 x^3$ is-

(A) 0 (B) 1
(C) 2 (D) 3

Graphs

Q28. Solve for x : $\log_{0.2}(x) < \log_{0.008}(27)$

(A) $x > 3$ (B) $x < 3$
(C) $x < 2$ (D) $x > 2$

Antilog

Q29. Antilog of $\frac{3}{2}$ to the base 81 is :-

(A) 829 (B) 729
(C) 730 (D) 725

Characteristic and Mantissa

Q30. Characteristics of the number $\frac{1}{27}$ when base of log is 2 will be :-

(A) 5 (B) –5
(C) 10 (D) –10

Q31. Number of integers whose charaterstics is 2 when base of log is 5 will be :

(A) 198 (B) 104
(C) 100 (D) 101

Q32. Number of digits in $(3)^{200}$:- ($\log 3 = 0.4771$)

(A) 98 (B) 96
(C) 99 (D) 95

Q33. Find the number of digits in $2^{12} \times 3^8$.

(A) 6 (B) 7
(C) 8 (D) 9

Q34. Number of cyphers (zeros) between decimal & first significant digit in $\left(\frac{1}{12}\right)^{100}$ is :-

(A) 102 (B) 104
(C) 107 (D) 110

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following when simplified reduces to an integer ?

(A) $\frac{2\log 6}{\log 12 + \log 3}$ (B) $\frac{\log 32}{\log 4}$

(C) $\frac{\log_5 16 - \log_5 4}{\log_5 128}$ (D) $\log_{1/4}\left(\frac{1}{16}\right)^2$

Q2. Which of the following when simplified, vanishes ?

(A) $\frac{1}{\log_3 2} + \frac{2}{\log_9 4} - \frac{3}{\log_{27} 8}$

(B) $\log_2\left(\frac{2}{3}\right) + \log_4\left(\frac{9}{4}\right)$

(C) $\log_8 \log_4 \log_2 16$

(D) $\log_{10} \cot 1° + \log_{10} \cot 2° + \log_{10} \cot 3° + \ldots\ldots + \log_{10} \cot 89°$

Q3. Let $N = \frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$. Then N is -

(A) a natural number (B) a prime number
(C) a rational number (D) an integer

Q4. $\frac{1}{1+\log_b a+\log_b c}+\frac{1}{1+\log_c a+\log_c b}+\frac{1}{1+\log_a b+\log_a c}$

is equal to-

(A) 1

(B) 2

(C) 10

(D) none of these

Q5. If a, b, c are distinct positive real numbers different from 1 such that $(\log_b a \cdot \log_c a - \log_a a) + (\log_a b \cdot \log_c b - \log_b b) + (\log_a c \cdot \log_b c - \log_c c) = 0$, then abc is equal to -

(A) 0

(B) e

(C) 1

(D) none of these

Q6. The solution set of the system of equations

$\log_3 x + \log_3 y = 2 + \log_3 2$ and $\log_{27}(x+y) = \frac{2}{3}$

is -

(A) (6, 3) (B) (3, 6)

(C) (6, 12) (D) (12, 6)

Q7. If $\log_{10} 3 = 0.477$, the no. of digits in 3^{40} is less than :

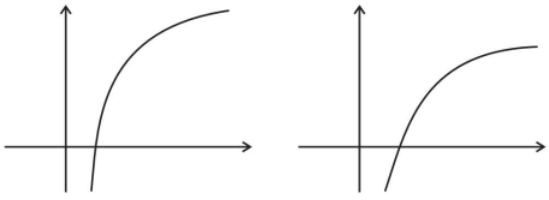
(A) 19 (B) 20

(C) 21 (D) 22

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # (Ques. 8 to 10)



$a > 1$ $a \in (0,1)$

Graph of $y = \log_a x$

If $a > 1$: log of greater number is greater.

If $a \in (0,1)$: log of greater number is smaller.

$\log_a N$ is positive if a & N lie on same side of unity while negative if a & N lie on opposite side of unity.

Q8. Identify the correct order :-

(A) $\log_2 6 < \log_3 8 < \log_3 6 < \log_4 6$

(B) $\log_2 6 > \log_3 8 > \log_3 6 > \log_4 6$

(C) $\log_3 8 > \log_2 6 > \log_3 6 > \log_4 6$

(D) $\log_3 8 > \log_4 6 > \log_3 6 > \log_2 6$

Q9. $\log_{\frac{1}{20}} 40$ is -

(A) greater than one

(B) smaller than one

(C) greater than zero and smaller than one

(D) none of these

Q10. $\log_{\frac{2}{3}} \frac{5}{6}$ is-

(A) less than zero

(B) greater than zero and less than one

(C) greater than one

(D) none of these

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11. **Column-I** **Column-II**

(i) $\frac{\log 2}{\log 4}$ (P) 1

(ii) $\frac{\log_5 16 - \log_5 4}{\log_5 128}$ (Q) 2

(iii) $\log_{\frac{1}{4}}\left(\frac{1}{16}\right)$ (R) $\frac{1}{2}$

(iv) $\frac{\log_{27} 8}{\log_3 2} =$ (S) $\frac{2}{7}$

(A) i → R ; ii → P ; iii → S ; iv → Q
(B) i → R; ii → S ; iii → Q ; iv → P
(C) i → S ; ii → Q ; iii → R ; iv → P
(D) i → S ; ii → P ; iii → Q ; iv → R

Q12. Match the column for values of x which satisfy the equation in Column-I

Column-I **Column-II**

(A) $\frac{\log_{10}(x-3)}{\log_{10}(x^2-21)} = \frac{1}{2}$ (P) 5

(B) $x^{\log x+4} = 32$, where base of logarithm is 2 (Q) 100

(C) $5^{\log x} - 3^{\log x-1} = 3^{\log x+1} - 5^{\log x-1}$ where the base of logarithm is 10 (R) 2

(D) $9^{1+\log x} - 3^{1+\log x} - 210 = 0$; where base of log is 3 (S) $\frac{1}{32}$

ASSERTION AND REASON

Q13. **Statement-I :**

If $\log_{(\log_5 x)} 5 = 2$, then $x = 5^{\sqrt{5}}$

Statement-II :

$\log_x a = b$, if $a > 0$, then $x = a^{1/b}$

(A) Statement-I is true, Statement-II is true; Statement-II is correct explanation for Statement-I.
(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for Statement-I.
(C) Statement-I is true, Statement-II is false.
(D) Statement-I is false, Statement-II is true.

SUBJECTIVE TYPE

Q14. The number $N = \frac{\log_5 250}{\log_{50} 5} - \frac{\log_5 10}{\log_{1250} 5}$ when simplified reduces to a natural number N. Then value of N is equal to-

Q15. Given $\log_3 a = p = \log_b c$ and $\log_b 9 = \frac{2}{p^2}$.

If $\log_9\left(\frac{a^4 b^3}{c}\right) = \alpha p^3 + \beta p^2 + \gamma p + \delta$ ($\forall$ p ∈ R – {0}), then ($\alpha+\beta+\gamma+\delta$) equals.

Q16. If $(\log_2 3).(\log_3 4).(\log_4 5)....\log_n(n+1) = 10$, then find the value of n.

Q17. Which is smaller ? 2 or $(\log_\pi 2 + \log_2 \pi)$.

Q18. Prove that $2^{\sqrt{\log_2 3}} = 3^{\sqrt{\log_3 2}}$.

Q19. If $a > 0; c > 0$; $b = \sqrt{ac}$; $ac \neq 1$

Prove that $\dfrac{\log_a N}{\log_c N} = \dfrac{\log_a N - \log_b N}{\log_b N - \log_c N}$

Q20. Compute the following :

(a) $\log_{1/3} \sqrt[4]{729.\sqrt[3]{9^{-1}.27^{-4/3}}}$

(b) $a^{\frac{\log_b(\log_b N)}{\log_b a}}$

Q21. Given that $\log_2 a = s$, $\log_4 b = s^2$ and

$\log_{c^2}(8) = \dfrac{2}{s^3+1}$.

Write $\log_2 \dfrac{a^2b^5}{c^4}$ as a function of 's'

(a,b,c>0, c≠1).

Q22. Prove that $a^x - b^y = 0$

where $x = \sqrt{\log_a b}$ & $y = \sqrt{\log_b a}$,

$a > 0$, $b > 0$ & $a, b \neq 1$.

Q23. Find the square of the sum of the roots of the equation

$\log_3 x.\log_4 x.\log_5 x = \log_3 x.\log_4 x + \log_4 x.\log_5 x + \log_5 x.\log_3 x$.

Q24. Solve for x :

(i) If $\log_4 \log_3 \log_2 x = 0$.

(ii) If $\log_e \log_5 \left[\sqrt{2x-2} + 3\right] = 0$

Q25. Solve : $\left(x^{\log_{10} 3}\right)^2 - 2\left(3^{\log_{10} x}\right) - 3 = 0$

Q26. Let $a > 1$ be a real number then solve

$a^{2\log_2 x} = 5 + 4x^{\log_2 a}$.

Q.27. Find the value of x satisfying

$\log_{10}(2^x + x - 41) = x(1 - \log_{10} 5)$.

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Properties of Logarithm

Q1. If $\log_a x = p$ and $\log_b x^2 = q$, then $\log_x \sqrt{ab}$ is equal to (where a, b, $x \in R^+ - \{1\}$) -

(A) $\dfrac{1}{p} + \dfrac{1}{q}$ (B) $\dfrac{1}{2p} + \dfrac{1}{q}$

(C) $\dfrac{1}{p} + \dfrac{1}{2q}$ (D) $\dfrac{1}{2p} + \dfrac{1}{2q}$

Q2. If $\log_{12} 27 = x$, then find $\log_6 16$ in terms of x :

(A) $4\left(\dfrac{3-x}{3+x}\right)$ (B) $4\left(\dfrac{3+x}{3-x}\right)$

(C) $8\left(\dfrac{3+x}{3-x}\right)$ (D) $\left(\dfrac{3+x}{3-x}\right)$

Q3. The ratio $\dfrac{2^{\log_{2^{1/4}} a} - 3^{\log_{27}(a^2+1)^3} - 2a}{7^{4\log_{49} a} - a - 1}$

simplifies to -

(A) $a^2 - a - 1$

(B) $a^2 + a - 1$

(C) $a^2 - a + 1$

(D) $a^2 + a + 1$

Q4. Value of expressions :

$4^{5\log_{4\sqrt{2}}\left(3-\sqrt{6}\right)-6\log_8\left(\sqrt{3}-\sqrt{2}\right)}$ is equal to -

(A) 8 (B) 9 (C) 10 (D) 11

Q5. Expression :

$$\frac{81^{\frac{1}{\log_5 9}}+3^{\frac{3}{\log_{\sqrt{6}}3}}}{409}\cdot\left(\left(\sqrt{7}\right)^{\frac{2}{\log_{25}7}}-(125)^{\log_{25}6}\right)$$

simplify to :

(A) 1 (B) 0 (C) 2 (D) 3

Q6. Let

$$y=\sqrt{\log_2 3.\log_2 12.\log_2 48.\log_2 192+16}-\log_2 12.\log_2 48+10.$$

Then $y \in N$ is :

(A) –6 (B) 6 (C) 7 (D) 5

Logarithm Equation

Q7. Given system of simultaneous equations $2^x.5^y=1$ and $5^{x+1}.2^y=2$. Then -

(A) $x=\log_{10}5$ and $y=\log_{10}2$

(B) $x=\log_{10}2$ and $y=\log_{10}5$

(C) $x=\log_{10}\left(\frac{1}{5}\right)$ and $y=\log_{10}2$

(D) $x=\log_{10}5$ and $y=\log_{10}\left(\frac{1}{2}\right)$

Q8. Solve :

$(\log_3 x)(\log_5 9)-\log_x(25)+\log_3 2=\log_3 54$

(A) $25,\frac{1}{\sqrt{5}}$ (B) $5,\frac{1}{\sqrt{5}}$

(C) $5,\frac{1}{25}$ (D) $5,\frac{1}{5}$

Q9. If $\log_{3/4}\log_8(x^2+7)+\log_{1/2}\log_{1/4}(x^2+7)^{-1}=-2$, then $|x|$ equals :

(A) 3 (B) –3 (C) –6 (D) 6

Q10. Solve : $\log_{(2x-1)}\left(\frac{x^4+2}{2x+1}\right)=1$ then x equal to

(A) –1 (B) 1

(C) $-\sqrt{3}$ (D) $\sqrt{3}$

Characteristic, Mantissa and Antilog

Q11. The greatest value of $(4\log_{10}x-\log_x(.0001))$ for $0<x<1$ is-

(A) 4 (B) –4 (C) 8 (D) –8

Q12. Which of the following statements is incorrect ?

(A) $7^{1/7}>(42)^{1/14}>1$

(B) $\log_3(5)\log_7(9)\log_{11}(13)>-2$

(C) $\sqrt{99+70\sqrt{2}}+\sqrt{99-70\sqrt{2}}$ is rational

(D) $\frac{1}{\log_4 3}+\frac{1}{\log_7 3}>3$

Q13. If $\left(\log_\beta\alpha\right)^2+\left(\log_\alpha\beta\right)^2=79$, $(\alpha>0,\beta>0,\alpha\neq1,\beta\neq1)$ then absolute value of $(\log_\beta\alpha)+(\log_\alpha\beta)$ can be-

(A) 7 (B) –9 (C) 9 (D) –7

Q14. Number of cyphers after decimal before a significant figure comes in $\left(\frac{5}{3}\right)^{-100}$ is -

(A) 21 (B) 22 (C) 23 (D) none

Q15. Let **'L'** denotes the antilog of 0.4 to the base 1024 and **'M'** denotes the number of digits in 6^{10} (Given $\log_{10}2=0.3010$, $\log_{10}3=0.4771$) and **'N'** denotes the number of positive integers which have the characteristic 2, when base of the logarithm is 6.

Then the value of LMN.

(A) 23041 (B) 23040

(C)23000 (D) 0

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is true?

(A) $7\log\frac{16}{15}-5\log\frac{25}{24}+3\log\frac{81}{80}=\log 2$

(B) $7\log\frac{16}{15}+5\log\frac{25}{24}+3\log\frac{81}{80}=\log 2$

(C) $5^{\log_7 3^2}+4^{\log_2 5}-9^{\log_7 5}=25$

(D) $5^{\log_7 3^2}+4^{\log_2 5}-9^{\log_7 5}=5$

Q2. Which of the following when simplified reduced to unity ?

(A) $\log_{1.5}\log_4\log_{\sqrt{3}}81$

(B) $\log_2\sqrt{6}+\log_2\sqrt{\frac{2}{3}}$

(C) $-\frac{1}{6}\log_{\frac{\sqrt{3}}{2}}\left(\frac{64}{27}\right)$

(D) $\log_{3.5}(1+2+3\div 6)$

Q3. Which of the following are correct ?

(A) $\log_3 19\,.\,\log_{1/7}3.\log_4 1/7>2$

(B) $\log_5(1/23)$ lies between -2 & -1

(C) if $m=4^{\log_4 7}$ and $n=\left(\frac{1}{9}\right)^{-2\log_3 7}$

then $n=m^4$

(D) $\log_{\sqrt{5}}\sin\left(\frac{\pi}{5}\right).\log_{\sqrt{\sin\frac{\pi}{5}}}5$ simplifies to an irrational number

Q4. In which of the following $(m>n)$ $(m,n\in R)$

(A) $m=(\log_2 5)^2$ and $n=\log_2 20$

(B) $m=\log_{10}2$ and $n=\log_{10}\sqrt[3]{10}$

(C) $m=\log_{10}5\cdot\log_{10}20+(\log_{10}2)^2$ and $n=1$

(D) $m=\log_{1/2}\left(\frac{1}{3}\right)$ and $n=\log_{1/3}\left(\frac{1}{2}\right)$

Q5. The equation

$(\log_x 10)^3-(\log_x 10)^2-6(\log_x 10)=0$ is satisfied by which values of x :

(A) $10^{1/3}$ (B) $10^{-1/2}$

(C) $10^{1/2}$ (D) $10^{-1/3}$

Q6. If $N=\left(\frac{1}{0.4}\right)^{20}$ then N contains n digits before decimal; where n is less than or equal to.

(A) 7 (B) 8

(C) 9 (D) 10

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 7 to 9)

Let $a^{\log_b x}=c$ where a,b,c & x are parameters.
On the basis of above information, answer the following questions :

Q7. If $a=2$, $x=a^{\log_5 25}$ & $c=\sqrt{2}$, then b is -

(A) 4 (B) 16

(C) 12 (D) 2

Q8. If $a = b = 2 \times 3^{\log_{10}\left(\sin^2\theta + \frac{1}{1+\tan^2\theta}\right)}$ &

$c = \frac{4\tan\theta}{\sqrt{\sec^2\theta - 1}}$ (where θ lies in Ist Quadrant),

then x is -

(A) 3 (B) –3

(C) 0 (D) 4

Q9. If $b = \log_{\sqrt{3}} 3, c = 2\log_b \sqrt{b}$ and $\sin\theta = a$ (where $x > 1$), then θ can be -

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{2}$

(C) $\frac{\pi}{2}$ (D) 0

Paragraph # 2 (Ques. 10 to 12)

Sometimes to solve an equation we may use the identity $a^{\log_a b} = b,\ b > 0,\ a > 0,\ a \neq 1$.

Q10. The number of solution of $x^{\log_x (x+3)^2} = 16$ is-

(A) 0 (B) 1

(C) 2 (D) More then 2

Q11. Value of x satisfying the equation $4x^{\log_2 \sqrt{x}} = x^2$ is-

(A) ϕ (B) 2

(C) 4 (D) 8

Q12. Solution set of $x^{\log_4 x} = 2^{3(\log_4 x + 3)} =$ is -

(A) {64, 1/8} (B) {64}

(C) {1/8} (D) ϕ

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13.

Column-I	**Column-II.**
(A) $3\log 4 + 2\log 5 - \frac{1}{3}\log 64 - \frac{1}{2}\log 16$ (where base of log is 10) equals:	(P) 2
(B) $2^{\log 3 - \log 5} \cdot 3^{\log 5 - \log 2} \cdot 5^{\log 2 - \log 3}$	(Q) 1
(C) Absolute value of solution of the equation $\sqrt{7^{2x^2-5x-6}} = \left(\sqrt{2}\right)^{3\log_2 49}$	(R) +3/2
	(S) 4
	(T) 5

Q14.

Column I	**Column II**
(A) If $\log_3 (5+8\log_{49}(5+4\log_{49}7)) = \lvert k\rvert$ then value of k is	(P) 0
(B) No. of roots of equation $\log_x (x^2 - 1) = 0$ is	(Q) 1
(C) If $x = \sqrt{9+\sqrt{77}}$ then value of $\frac{1}{11}\left(x + \frac{2}{x}\right)^2$ is	(R) 2
(D) $\sqrt{(\log_{0.5} 4)^2}$	(S) –2

SUBJECTIVE TYPE

Q15. If $a = \log_{12}18$ & $b = \log_{24}54$ then find the value of $ab + 5(a - b)$.

Q16. Given that $\log_2 3 = a$, $\log_3 5 = b$ and $\log_7 2 = c$, express the logarithm of the number 63 to the base 140 in terms of a,b & c.

Q17. Find all real numbers x which satisfy the equation, $2\log_2 \log_2 x + \log_{1/2} \log_2 (2\sqrt{2}x) = 1$.

Q18. Let $\log_2 x + \log_4 y + \log_4 z = 2$

$\log_9 x + \log_3 y + \log_9 z = 2$

and $\log_{16} x + \log_{16} y + \log_4 z = 2$.

Find the value of $\frac{yz}{x}$.

Q19. Let a and b be real numbers greater than 1 for which there exists a positive real number c, different from 1, such that $2(\log_a c + \log_b c) = 9\log_{ab} c$. Find the largest possible value of $\log_a b$.

Q20. If a,b and c are positive real numbers such that $a^{\log_3 7} = 27; b^{\log_7 11} = 49$ and $c^{\log_{11} 25} = \sqrt{11}$. Find the value of

$\left(a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}\right)$.

Q21. Find the product of the positive roots of the equation $\sqrt{(2017)}(x)^{\log_{2017} x} = x^2$.

Q22. If the product of the roots of the equation, $x^{\left(\frac{3}{4}\right)(\log_2 x)^2 + \log_2 x - \left(\frac{5}{4}\right)} = \sqrt{2}$ is $\frac{1}{\sqrt[b]{a}}$.

(where $a, b \in N$) then the least value of $(a + b)$.

Q23. Solve following equation

$x^{\frac{\log x + 5}{3}} = 10^{5 + \log x}$.

Q24. If 'x' and 'y' are real numbers such that $2\log(2y - 3x) = \log x + \log y$, find $\frac{x}{y}$.

Answer Key

Ex - 1									
1	2	3	4	5	6	7	8	9	10
B	C	D	D	A	A	A	C	C	C
11	12	13	14	15	16	17	18	19	20
A	C	B	B	D	A	A	A	C	C
21	22	23	24	25	26	27	28	29	30
D	A	B	B	A	A	B	A	B	B
31	32	33	34						
C	B	C	C						

Ex - 1A

1. A.D **2.** A.B.C,D **3.** A,B,C,D **4.** A **5.** C

6. A,B **7.** C,D **8.** B **9.** B **10.** B

11. B **12.** A → P; B → (R,S) ; C → Q ; D → P

13. A **14.** 2 **15.** 3 **16.** 1023 **17.** 2

20. (a) –1 **(b)** $\log_b N$ **21.** $2s + 10s^2 - 3(s^3 + 1)$ **23.** 3721

24. (i) 8 **(ii)** x= 3 **25. x = 10** **26.** $x = 5^{\log_a 2}$ **27.** 41

Ex - 2

1	2	3	4	5	6	7	8	9	10
D	A	D	B	A	B	C	A	A	D
11	12	13	14	15					
D	C	C	B	B					

Ex - 2A

1. B,C **2.** A,B,C,D **3.** A.B,C **4.** A,D **5.** A,B

6. B,C,D **7.** B **8.** D **9.** C **10.** A

11. C **12.** A

13. (A)→ P ; (B)→ Q ; (C)→ R, S

14. (A) → R, S ; (B) → Q ; (C) → R ; (D) → R **15. 1** **16.** $\frac{1+2ac}{2c+abc+1}$

17. 8 **18.** 54 **19.** 2 **20.** 469 **21.** $(2017)^2$

22. 19 **23.** $x = 10^3$, $x = 10^{-5}$ **24.** 4/9

-: Important Notes & Formulas :-